A brief note: at the time, this was our first foray into the use of quantum dots, and all of us were learning how to work with quantum dots, so this was a complication.

1. Using ChatGPT (GPT-4o) I learned a lot about quantum dots
    a. I hadn't expected to do a chemistry/QD project, since my original mentor (doing a cancer bioinformatics project) had an unexpected schedule conflict and had bowed out of the mentoring program. At the time, my quantum physics knowledge was effectively "entering USNCO" level. So, when I learned I was going to be doing research into quantum dots, I basically self-taught myself the basics of quantum dots using ChatGPT, and critically, I realized that given the particle-in-a-box model, you could control fluorescence using size, or, given base material properties (which was a bit of a big ask, but with reasonable assumptions wouldn't be too hard), estimate size using fluorescence. This would come into play later

2. Early on, found paper after low quantum yield and inability for red QDs
    a. At the time, our first synthesis had unexpectedly low quantum yield, and apparently very low thermostability (it wouldn't get larger than what I would later estimate to be 2-3 NM, so it would only produce green QDs). While green QDs would marginally work, the quantum yield was abysmal (estimated to be ~10-20 percent) and not good enough for a lateral flow assay (the proposed means of experiment at the time), and the concentration was probably not very high (although I wasn't able to make an estimate). So, I decided to find a few papers. After figuring out that coating QDs with a second layer (ZnS) would raise quantum yield and amine ligands could stabilize QDs enough to make them large enough for red/orange emissions, I decided to utilize a paper describing an experimental procedure satisfying these constraints (amine/phosphine ligand combo + coated with ZnS), and my mentor and I decided to go forwards with that paper
    b. Impact: At the time, we were unable to move forwards after unexpectedly low quantum yield and an unexpectedly short wavelength. My paper was accepted, and it allowed our research group to move on more effectively and successfully synthesize quantum dots of sufficient quantum yield and a long enough fluorescence wavelength.

3. Bioconjugation papers when pH controls didn't work
    a. Early on, using a synthetic peptide, a lab partner had been reporting excess non-specific bioconjugation, including on an essential lysine necessary for cleavage (which, in turn, was necessary for detecting botulinum toxin). Using EDC-NHS for bioconjugation basically modifies all -NH2 groups. Normally, this can be controlled by setting the pH of the solution; Lysines are more basic (have

a higher pKa), so with the right pH, the lysines can be protonated, thus making them less nucleophilic, and thus much less reactive to EDC-NHS.

b. However, this didn't work. Despite multiple trials at various pH values around the pKa of lysine, nonspecific bioconjugation kept occurring. At this time, I volunteered to do a paper search. The idea was to find papers that would allow N-terminal specific bioconjugation without too much pH control. I used my understanding of chemistry to determine which reactions were probably dependent on pH control for N-terminal specific bioconjugation, which included analyzing reaction mechanisms, graphs, and reading text descriptions. At the end of 72 hours, I had a set of papers, which I sent to my mentor. My PI actually said he'd use one of these papers for bioconjugation experiments in future, so I basically left a long term contribution with the papers I had selected.

c. Impact: At this time, bioconjugation was an impediment; without a sufficiently selective N-terminal bioconjugation method we'd have to use expensive synthetic peptides with non-canonical amino acids. Since I had an N-terminal specific bioconjugation method that didn't require pH controls, the project could move forwards without more synthetic peptides and we could bioconjugate peptides to quantum dots.

4. During mentor absence, decided to reuse mutagenesis experiment pentamer (originally had been used for mass spec) for bioconjugation
    a. Around this time, given mass spectrometry data, the synthetic peptide we had been using long-term as a backup for bioconjugation was beginning to degrade, and we had only a few micrograms left for the whole lab. Synthesis was pretty expensive and, given the large amount of testing we expected at the time, we were going to burn through the sample fairly quickly. However, we had previously been using a peptide pentamer using only the 20 canonical amino acids, which could be synthesized by E. coli. This had been used for mutagenesis experiments to determine which amino acid residues were actually necessary for cleavage by botulinum toxin catalytic submit
    b. Unlike the backup synthetic peptide, which could use maleimide (through Cys) or azide click chemistry (through a non-canonical amino acid) for bioconjugation (which wouldn't have the non-specific bioconjugation issue), this did not have a means of easy bioconjugation; the only way for bioconjugation was through the N-terminus (a variant with a few n-terminal cysteines apparently didn't get cleaved by the botulinum toxin catalytic subunit, given electrophoresis and western blot data). So, reusing that peptide would have required EDC-NHS, so that proposal was shot down
    c. However, I noticed that this peptide was available, there were still plasmids for it around, and given my research into N-terminal bioconjugation, I had papers to actually do it. So I decided to chat with my PI and several other coworkers, mentioning that I had papers that might actually sidestep the EDC-NHS problem, and that since we did have gene expression capacities, this might actually work. Later on, a backup plan to utilize the pentamer peptide I had proposed was

initiated, and I was involved in electrophoresis/western blot experiments for expressing the peptide while my mentor was gone

d. **Impact**: This saved our project a lot of space and time. Originally, we'd have to use a synthetic peptide (with resultant high costs). After I got the N-terminal bioconjugation methods, the idea of reusing an existing peptide (used for testing mutagenesis) became feasible (since now, there was no longer a need for non-canonical amino acids for bioconjugation), and since that could be made in bacteria at low cost, that saved the project a lot of time and money.

5. Used fluorescence data (Brus equation/"particle in a box" model) to estimate size at a time where TEM/DLS were not available
    a. While TEM/DLS are basically the gold standard for this experiment, the DLS experiment at our lab was not capable of measuring particles at the expected size, and the TEM specialist was on vacation. Since we needed to coat quantum dots, we needed an estimate of radius for A, estimating molar mass (because the paper required moles of nanoparticles) and B, estimating total area/mole (to estimate how much Zn and S precursor we needed for coating).
    b. Given my research at the beginning of the project, I figured that, given the "particle in a box model", I could back calculate the size of the quantum dot given the fluorescence data. So, I decided to take multiple fluorescence measurements given solvent, average them out, and then estimate the size of the quantum dot, for later use. After a few days, my PI approved the use of the estimate if the TEM specialist didn't come back in another week (well, the TEM specialist came back in time). Later, when data from TEM was acquired, I got an estimate within 2 nm of the size.
    c. **Impact**: This would have significantly sped up the project if it had been greenlighted. Since I had an estimate that was sufficiently close enough to TEM measurements, we would've had an accurate size measurement that would have allowed us to start the second phase of quantum dot synthesis earlier. If it had been greenlighted, it would've sped up the project by ~1-2 weeks.

6. Estimated the Beer-Lambert constant from size and density (used that for conc estimate, then got absorbance at max wavelength, and figured out Beer-Lambert constant)
    a. Normally, to measure the concentration, you'd use the Beer-Lambert's Law. Unfortunately, the research paper had not published the Beer-Lambert's constant, meaning we couldn't do that, and for future, we'd need that constant. So, I decided to do it in reverse: estimate concentration, and find out Beer-Lambert's constant. So, given the radius estimates (corrected, after TEM data came back), I estimate the molar mass of the CdSe/ZnS quantum dots (which wasn't very hard, given known densities of CdSe/ZnS), estimated the concentration of a hexane solution of CdSe/ZnS given density, and then used that to measure the Beer-Lambert's constant. Given that, we had concentration and Beer-Lambert's constant to reuse later.

   b. **Impact**: This was also necessary for coating the quantum dots with ZnS, and for measuring concentrations long term. By making that decision, I made the continuation of the project possible (we could actually coat the quantum dots with ZnS, allowing the project to move forwards), and long term, I'd say the project would be able to measure the concentration of quantum dots long term using spectroscopy (which is a very quick and easy process).

7. Confirmed synthesized QDs could, in fact, be a drop in replacement for Thermo-Fisher QDs
   a. At the time, the synthesis of the QDs was not quite finished and the end was beginning to approach. So, I decided to make the jump and estimate how useful the QDs would be for the future, as a contribution to the lab. To do so, I measured the absorption spectra of synthesized and Thermo-Fisher QDS (what we were using at the time) and estimated the quantum yield and Beer-Lambert's constant (given the concentration estimates) Although I didn't publish this data on my poster as I had long since run out of space, I did leave these notes to my lab, and my PI mentioned she'd be using this later.
   b. **Impact:** This was kind of the point of the project, and by confirming near-identical spectral properties, we could move forwards with in house synthesis.

8. Confirmed that chloroform was likely cause of anomalous fluorescence loss; found a set of papers to figure out a solution
   a. At the end, we were originally going to functionalize the quantum dots to solubilize them in H_2O; a project that had originally been planned for 3 weeks before it had happened. There was a final delay. The original procedure required extraction via chloroform, but when my PI and I attempted an extraction procedure, it ended up losing fluorescence. Via a few tests (mixing QDs with various solvents and measuring fluorescence losses, and then testing after centrifugation), I confirmed that the QDs were very sensitive to chloroform.
   b. My mentor had a few other extraction procedures. However, the functionalization procedure we had found required the use of chloroform. So, I decided to find a few procedures. The constraints were fairly tight: not using chloroform, no use of high-temperature equipment, and only using stuff that we had at the time. So, I found a paper, and sent it over to my mentor. We would have used that, but time ran out. However, my mentor said she'd actually use it long term.
   c. **Impact:** by finding a non-chloroform procedure and confirming that chloroform was likely the cause of fluorescence loss, I made long term progress on QD functionalization/water solubilization possible.